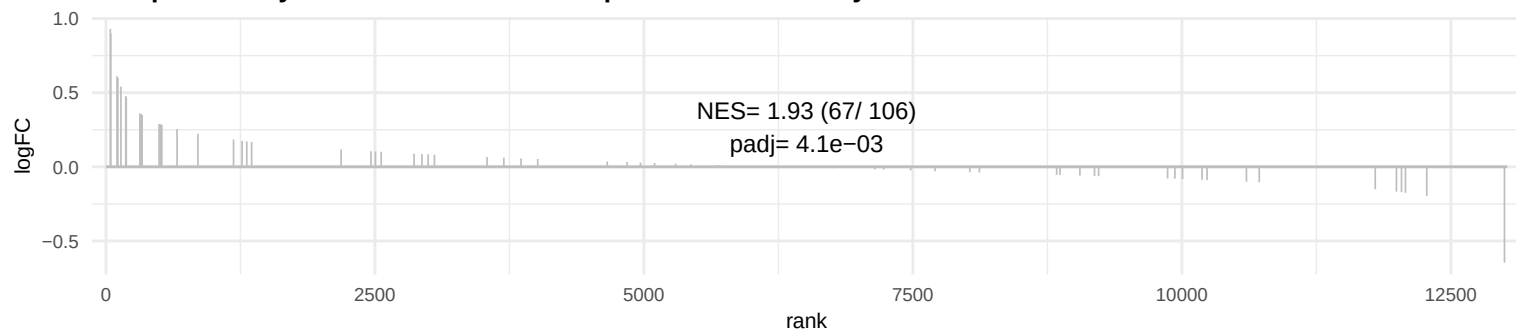


Complement System In Neuronal Development And Plasticity WP5090

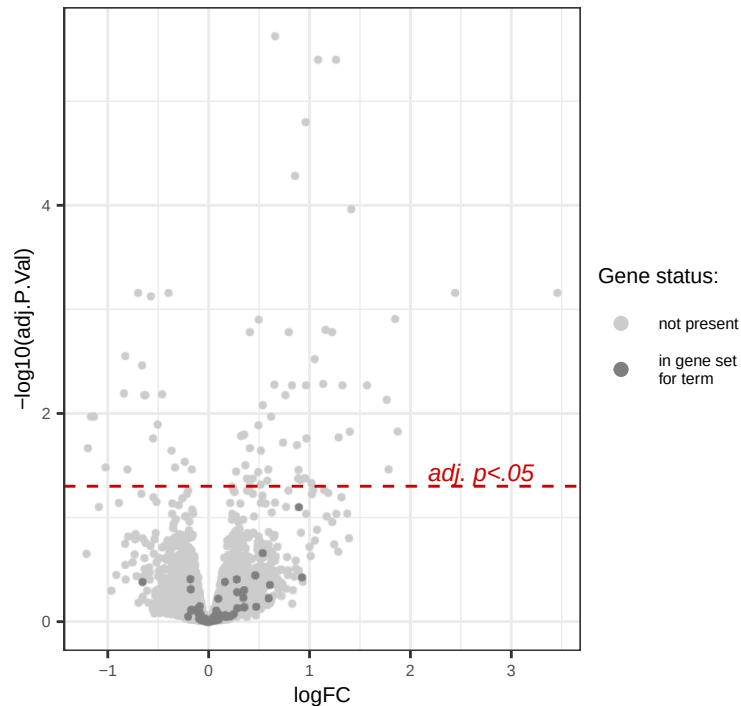


Since fgsea results are calculated based on the ranked logFC values, many fgsea results with $\text{padj} < .05$ do not contain any L-R DEGs. For our analysis, we have filtered out such results.

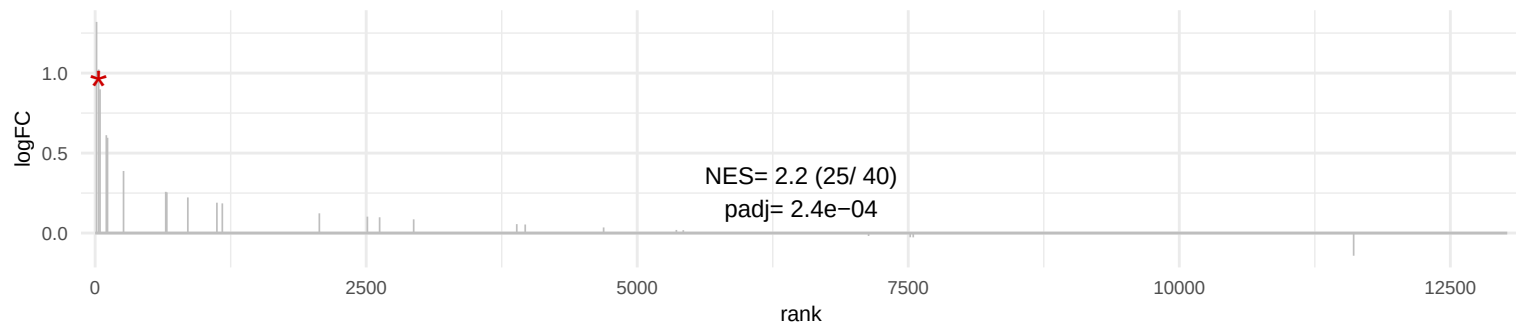
fgsea manhattan plot (top): Lowly ranked values (x-axis) have the highest, most positive logFC (y-axis).

Volcano plot (right): Opposite from above, the logFC (x-axis) shows negative values on left and positive values on right.

Volcano showing genes contributing to GSEA result



Microglia Pathogen Phagocytosis Pathway WP3937

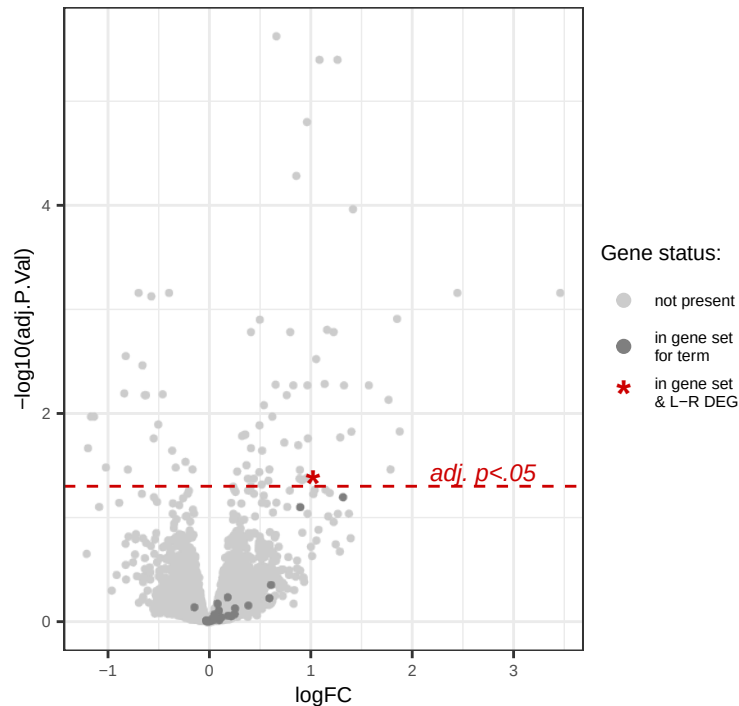


Significant ($\text{padj} < .05$) fgsea results were filtered to those that contained a L-R DEG in the leading edge genes. These L-R DEGs did not have to be paired (sig. in both annotations), allowing for unique PRECAST cluster or Seurat cell type label results to emerge (if present).

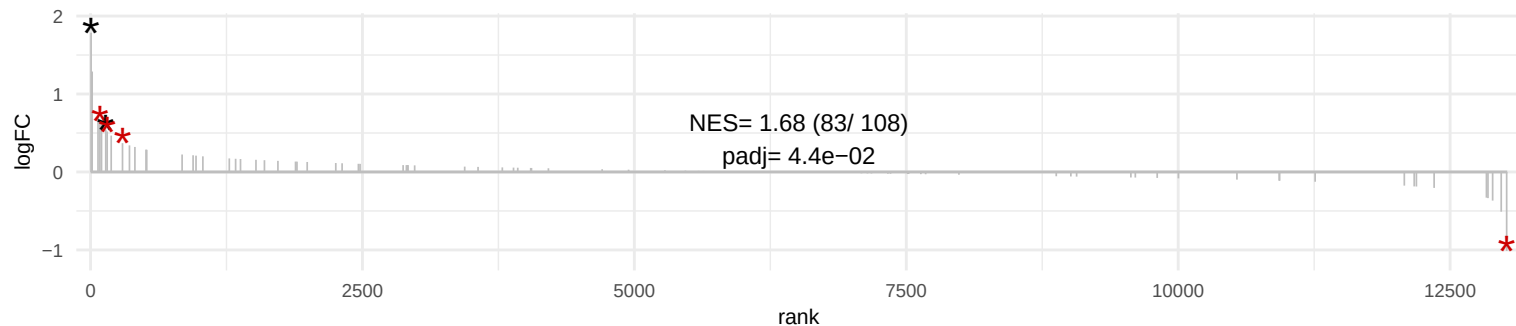
fgsea manhattan plot (top): The rank position of L-R DEGs is indicated with a red asterisk.

Volcano plot (right): The L-R DEGs are indicated with a red asterisk.

Volcano showing genes contributing to GSEA result



Macrophage Stimulating Protein MSP Signaling WP5353

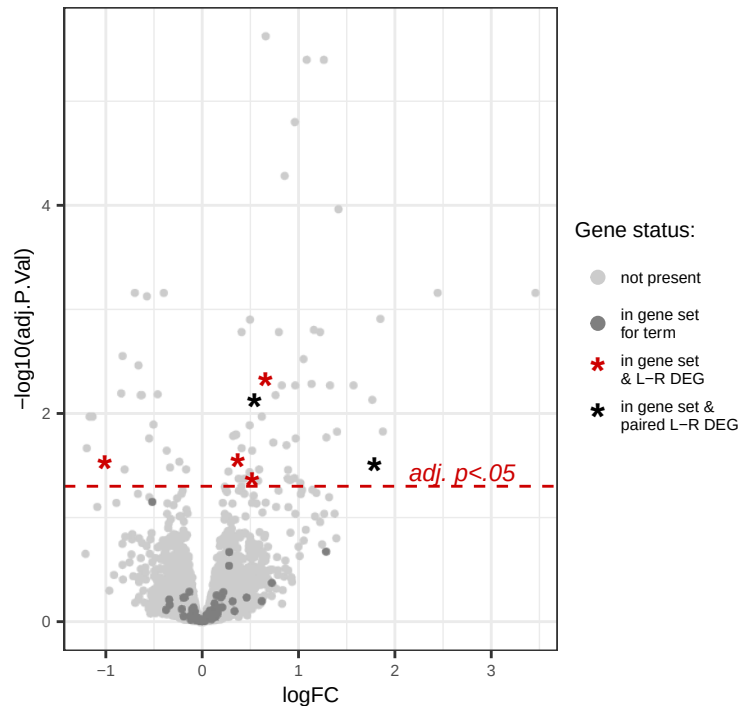


Significant ($\text{padj} < .05$) fgsea results containing a L-R DEG were further stratified by those results that also contain a paired L-R DEG in the leading edge genes. This will help distinguish the smaller number of sig. results that only contained layer-specific L-R DEGs that were not reproducible in the other annotation strategy.

fgsea manhattan plot (top): The rank position of paired L-R DEGs is indicated with a black asterisk. Any additional layer-specific L-R DEGs are also indicated with a red asterisk.

Volcano plot (right): As this example shows, it is possible for a L-R DEG to be significantly different in the opposite direction of the GSEA result.

Volcano showing genes contributing to GSEA result



NTC.MDD F WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	74	68	58	56	53
smoothed	L2_L3_L4	L3.4	65	55	39	56	38
seurat	L2_L3_L4	L2.3	67	80	49	56	49
seurat	L2_L3_L4	L4	52	61	33	56	33
smoothed	L5	L5	83	51	57	44	57
seurat	L5	L5	63	68	49	44	40
smoothed	L6	L6	78	60	60	46	59
seurat	L6	L6	68	68	57	46	54
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	74	131	66		
smoothed	L2_L3_L4	L3.4	65	131	58		
seurat	L2_L3_L4	L2.3	67	131	61		
seurat	L2_L3_L4	L4	52	131	47		
smoothed	L5	L5	83	131	75		
seurat	L5	L5	63	131	56		
smoothed	L6	L6	78	131	69		
seurat	L6	L6	68	131	60		

NTC.MDD M WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	14	0	0	0	0
smoothed	L2_L3_L4	L3.4	78	0	0	0	0
seurat	L2_L3_L4	L2.3	22	0	0	0	0
seurat	L2_L3_L4	L4	13	0	0	0	0
smoothed	L5	L5	72	0	0	0	0
seurat	L5	L5	93	0	0	0	0
smoothed	L6	L6	59	0	0	0	0
seurat	L6	L6	60	0	0	0	0
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	14	0	0		
smoothed	L2_L3_L4	L3.4	78	0	0		
seurat	L2_L3_L4	L2.3	22	0	0		
seurat	L2_L3_L4	L4	13	0	0		
smoothed	L5	L5	72	0	0		
seurat	L5	L5	93	0	0		
smoothed	L6	L6	59	0	0		
seurat	L6	L6	60	0	0		

NTC.BPD F WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	8	23	2	21	2
smoothed	L2_L3_L4	L3.4	11	12	2	21	2
seurat	L2_L3_L4	L2.3	11	32	4	21	1
seurat	L2_L3_L4	L4	8	16	0	21	0
smoothed	L5	L5	15	32	5	26	5
seurat	L5	L5	12	41	5	26	5
smoothed	L6	L6	5	77	2	42	1
seurat	L6	L6	16	56	8	42	8
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	8	73	5		
smoothed	L2_L3_L4	L3.4	11	73	7		
seurat	L2_L3_L4	L2.3	11	73	7		
seurat	L2_L3_L4	L4	8	73	4		
smoothed	L5	L5	15	73	7		
seurat	L5	L5	12	73	7		
smoothed	L6	L6	5	73	2		
seurat	L6	L6	16	73	11		

NTC.BPD M WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	64	51	43	34	35
smoothed	L2_L3_L4	L3.4	63	38	42	34	39
seurat	L2_L3_L4	L2.3	69	44	52	34	41
seurat	L2_L3_L4	L4	8	9	3	34	3
smoothed	L5	L5	40	28	19	17	18
seurat	L5	L5	51	23	22	17	22
smoothed	L6	L6	33	15	16	8	16
seurat	L6	L6	26	13	15	8	15
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	64	64	48		
smoothed	L2_L3_L4	L3.4	63	64	52		
seurat	L2_L3_L4	L2.3	69	64	57		
seurat	L2_L3_L4	L4	8	64	7		
smoothed	L5	L5	40	64	35		
seurat	L5	L5	51	64	44		
smoothed	L6	L6	33	64	27		
seurat	L6	L6	26	64	20		

MDD.BPD F WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	50	11	20	9	19
smoothed	L2_L3_L4	L3.4	49	3	13	9	13
seurat	L2_L3_L4	L2.3	37	13	20	9	20
seurat	L2_L3_L4	L4	40	3	12	9	12
smoothed	L5	L5	29	0	0	0	0
seurat	L5	L5	25	4	4	0	0
smoothed	L6	L6	23	11	4	8	4
seurat	L6	L6	16	12	6	8	6
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	50	20	28		
smoothed	L2_L3_L4	L3.4	49	20	27		
seurat	L2_L3_L4	L2.3	37	20	25		
seurat	L2_L3_L4	L4	40	20	23		
smoothed	L5	L5	29	20	17		
seurat	L5	L5	25	20	14		
smoothed	L6	L6	23	20	14		
seurat	L6	L6	16	20	14		

MDD.BPD M WikiPathways

annotation	neuron group	source	# sig. terms	# LR DEGs	# sig. terms with LR DEG	# paired LR degs	# sig. terms with paired LR DEG
smoothed	L2_L3_L4	L2	108	28	36	38	26
smoothed	L2_L3_L4	L3.4	132	44	84	38	77
seurat	L2_L3_L4	L2.3	126	46	78	38	75
seurat	L2_L3_L4	L4	49	62	24	38	19
smoothed	L5	L5	111	18	25	14	25
seurat	L5	L5	162	33	50	14	26
smoothed	L6	L6	109	2	14	2	14
seurat	L6	L6	110	13	20	2	13
annotation	neuron group	source	# sig. terms	# LA & LR DEGs	# sig. terms with LA & LR DEG		
smoothed	L2_L3_L4	L2	108	78	90		
smoothed	L2_L3_L4	L3.4	132	78	113		
seurat	L2_L3_L4	L2.3	126	78	111		
seurat	L2_L3_L4	L4	49	78	44		
smoothed	L5	L5	111	78	95		
seurat	L5	L5	162	78	125		
smoothed	L6	L6	109	78	93		
seurat	L6	L6	110	78	94		